CALIBRATION QUALITY REPORT

Camera calibration diagnostic

FULL REPORT · intrinsics, coverage and uncertainty

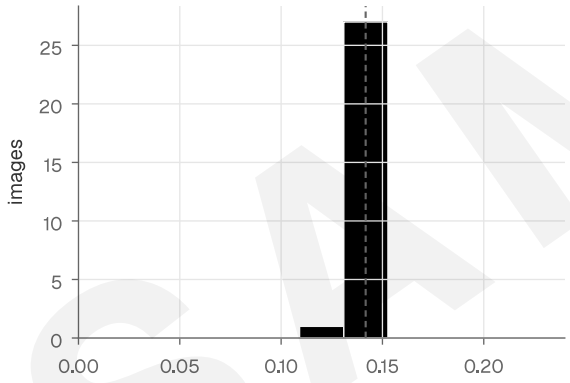
CALIBRATION GRADE		
B	**Conditioning requires additional coverage.** RMS 0.142 px · coverage 66.2 % · 2 views rejected before final solve.	28 / 30 used 2448 × 2048 px

1 Inputs

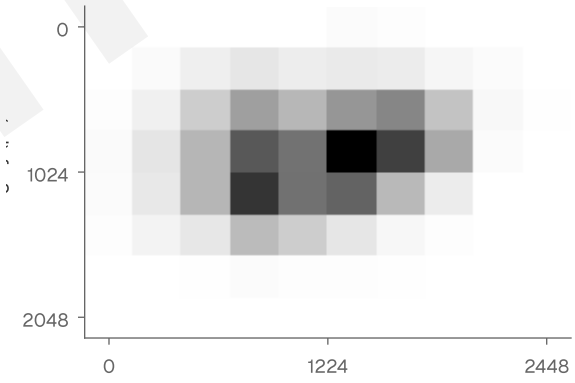
Camera model	[User camera] 5 MP GigE, 2448 × 2048
Pixel pitch / lens	3.45 µm / 12 mm
Images supplied	30 · 28 used · 2 rejected
Target	CT-CHA-C-200150-6X6 · SN 000311
Target inspection data	CT-IR-000311 imported
Software	OpenCV 5.0.0 · calibrateCameraExtended

RMS REPROJECTION	MAX PER-IMAGE	COVERED GRID	REJECTED
0.142 px	0.152 px	66.2 %	2 views

2 Summary diagnostic



Histogram · retained per-image RMS. Dashed line: final RMS.



Coverage heatmap · 28 retained views; black = more detected corners.

3 Recommendations

- Recapture rejected views 09, 24 with focus locked; do not include them in the next solve.
- Add three captures with the target projected into the lower-right image region, at 0.45–0.80 m and 20–35° tilt.
- Retain the current centre/edge range and repeat the solve after the coverage grid exceeds 75 % for grade A.

GRADE RUBRIC A RMS ≤ 0.20 px, ≥ 75 % grid, ≤ 2 rejected · B RMS ≤ 0.30 px, ≥ 55 %, ≤ 3 · C RMS ≤ 0.50 px, ≥ 35 %, ≤ 5 · D otherwise. All three conditions apply within each grade.

SAMPLE

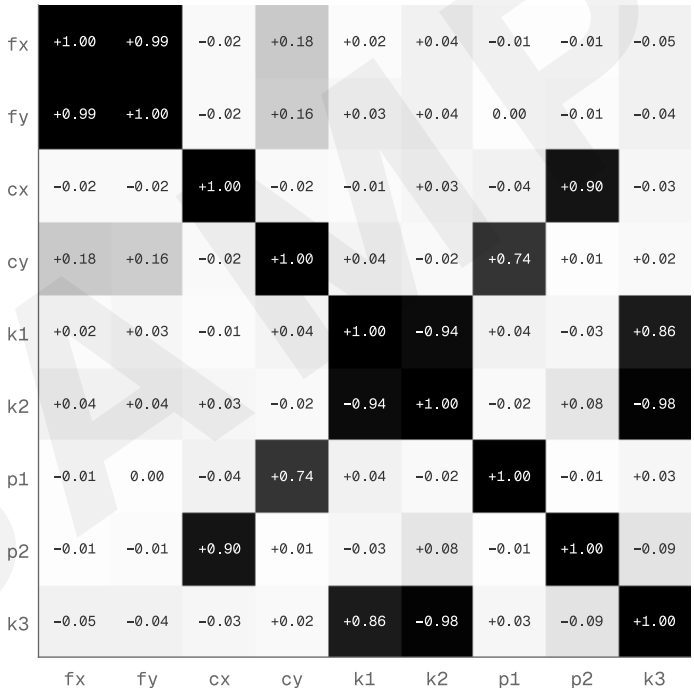
4 Intrinsic parameters

final solve · 28 retained views

Estimates and standard deviations below are produced by the final OpenCV solve. The covariance uses the projectPoints Jacobian with per-view poses eliminated.

PARAMETER	ESTIMATE	STD. DEVIATION	UNIT
fx	3479.860	0.74322	px
fy	3476.612	0.73141	px
cx	1226.752	0.98543	px
cy	1027.455	0.77875	px
k1	−0.12367	0.00139	—
k2	0.14795	0.02866	—
p1	0.00063	0.00005	—
p2	−0.00055	0.00006	—
k3	−0.37733	0.17989	—

5 Intrinsic-parameter correlation



Grey heatmap of the covariance-derived correlation matrix. Dark cells indicate strong positive or negative correlation; signs are printed in each cell.

METHOD

Residual variance = 0.01023 px² · degrees of freedom = 9679. Coupling is expected when an image region is sparsely covered.

SAMPLE

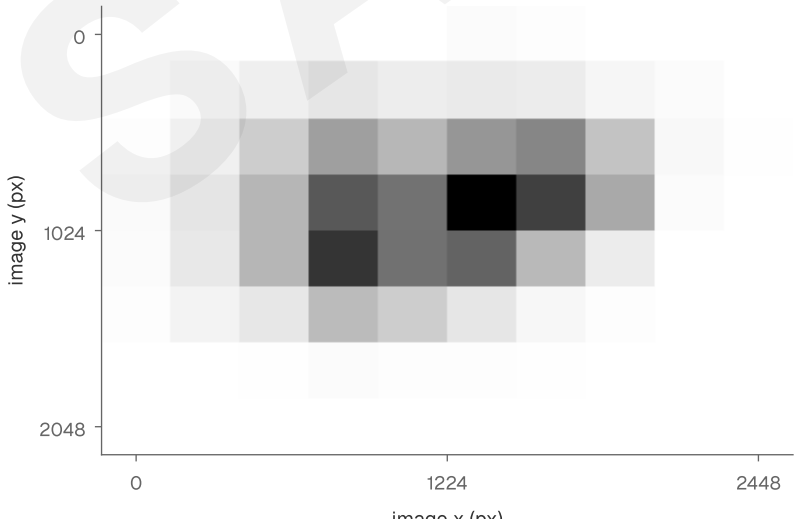
6 Per-image results

30 supplied views

Final solve retained 28 views. Views 09, 24 were excluded after the preliminary all-view error screen.

IMAGE	ERROR (px)	DISTANCE (mm)	TILT (°)	DISPOSITION
01	0.130	523	5.0	used
02	0.131	425	38.6	used
03	0.147	550	26.3	used
04	0.144	626	18.6	used
05	0.146	726	18.8	used
06	0.142	870	8.6	used
07	0.140	528	37.1	used
08	0.152	765	5.9	used
09	1.026	584	19.6	excluded
10	0.131	604	8.0	used
11	0.136	882	22.9	used
12	0.141	606	24.6	used
13	0.139	880	37.2	used
14	0.142	853	24.7	used
15	0.140	596	30.9	used
16	0.140	807	8.7	used
17	0.136	825	16.7	used
18	0.148	490	7.7	used
19	0.141	748	28.5	used
20	0.132	881	31.8	used
21	0.142	511	35.7	used
22	0.139	409	6.0	used
23	0.152	845	16.5	used
24	1.081	886	6.4	excluded
25	0.142	899	17.5	used
26	0.146	684	14.3	used
27	0.148	898	9.0	used
28	0.147	647	24.5	used
29	0.149	769	6.0	used
30	0.143	811	22.4	used

7 Coverage detail



Detected ChArUco corners per image-plane cell; retained views only.

REJECTED VIEWS

09, 24 · blur / low contrast; corner outliers

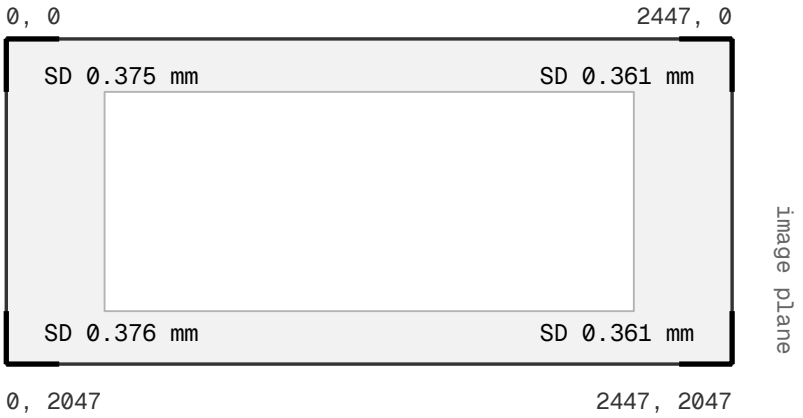
COVERAGE READING

66.2 % of the 8 × 10 grid contains retained detections. The lower-right region is sparse by design.

SAMPLE

8 Uncertainty at a 600 mm working distance

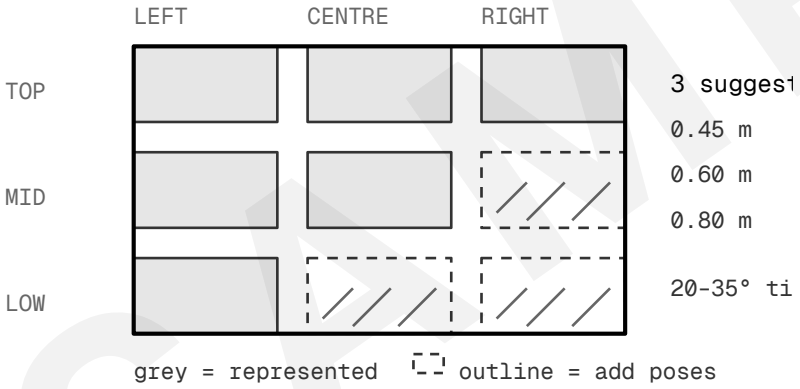
Monte Carlo propagation uses 6000 draws from the intrinsic covariance. Values are one-sigma positional uncertainty after inverse projection to a plane at 600 mm.



WORKING-PLANE MODEL
Image pixels are undistorted with each sampled intrinsic parameter set, then back-projected to the 600 mm plane.

IMAGE CORNER	PIXEL COORDINATE	SD POSITION (mm)	MC BIAS (mm)
top-left	0 × 0	0.3749	0.0047
top-right	2447 × 0	0.3614	0.0028
bottom-left	0 × 2047	0.3760	0.0039
bottom-right	2447 × 2047	0.3614	0.0023

9 Missing views



SUGGESTED CAPTURE SET
Place the board in each outlined region. Keep full corner detection and vary distance as indicated.

10 Target contribution

Inspection data for CT-CHA-C-200150-6X6 SN 000311 was imported from CT-IR-000311. Equivalent pixel values use the final mean focal length 3478.2 px at 600 mm.

TARGET CONTRIBUTION	BOARD RESULT	UNIT	AT 600 MM	COMPARISON
IR RMS board deviation	0.65	µm	0.0038 px	2.7 % of RMS
IR maximum board deviation	1.6	µm	0.0093 px	6.5 % of RMS

11 Statement

Computed from user-supplied data; Calib.Tools has not verified the images or the camera.

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SAMPLE